



SC1006

Computer Organisation and Architecture

Computer System Overview

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Part 2 of this course

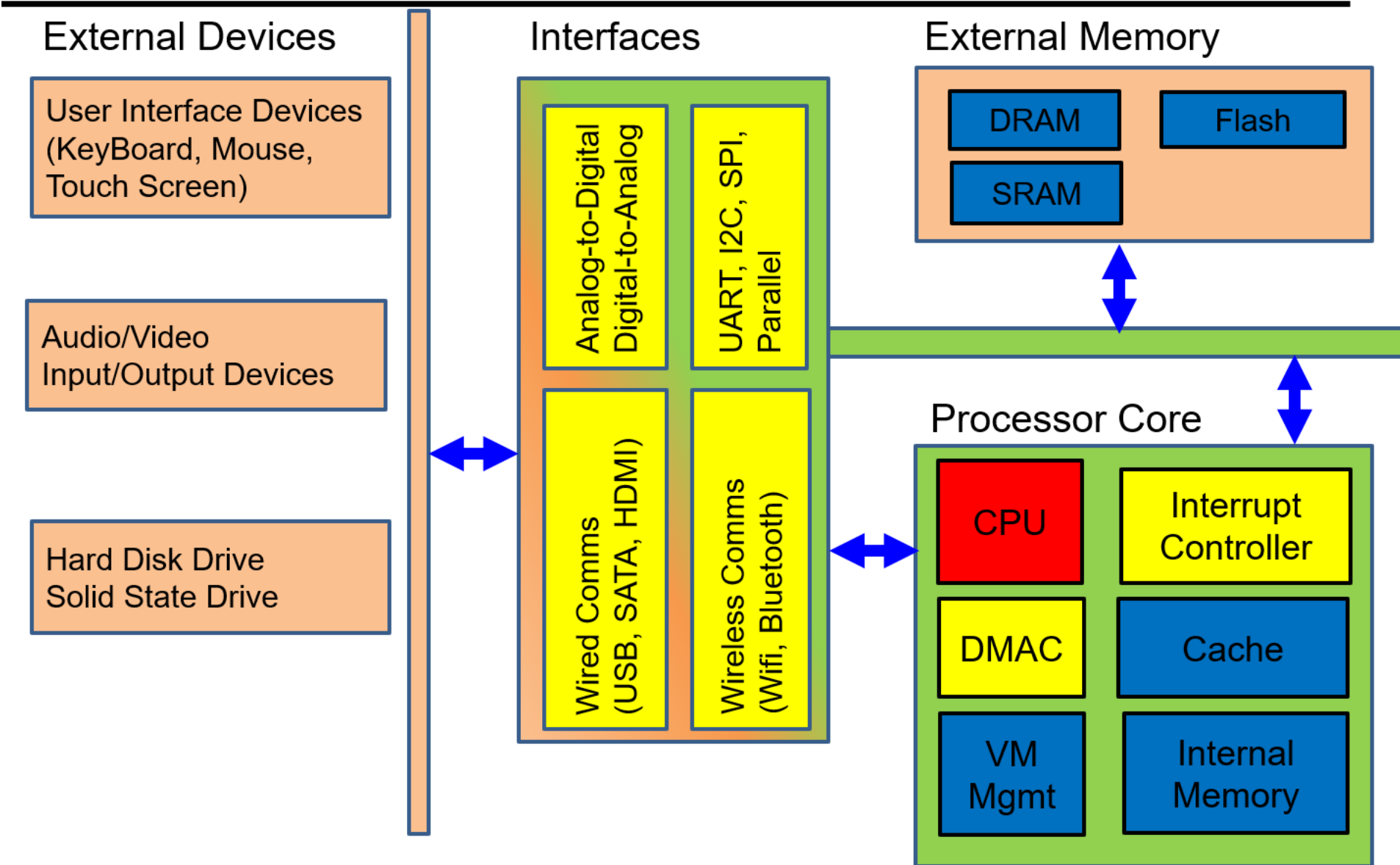
- “**Flipped Classroom**” mode
 - ❑ Lecture content pre-recorded on-line, with MCQs for self-assessment
 - ❑ 2-hr physical session per week: key content summary, further practice and consultation
- Challenges of Part 2 content (in comparison with Part 1)
 - ❑ **More topics**
 - ❑ **Wide content diversity by nature**
- Some topics to be further covered in advanced courses
 - ❑ They are only briefly introduced in this course

Let me introduce Part 2 contents from

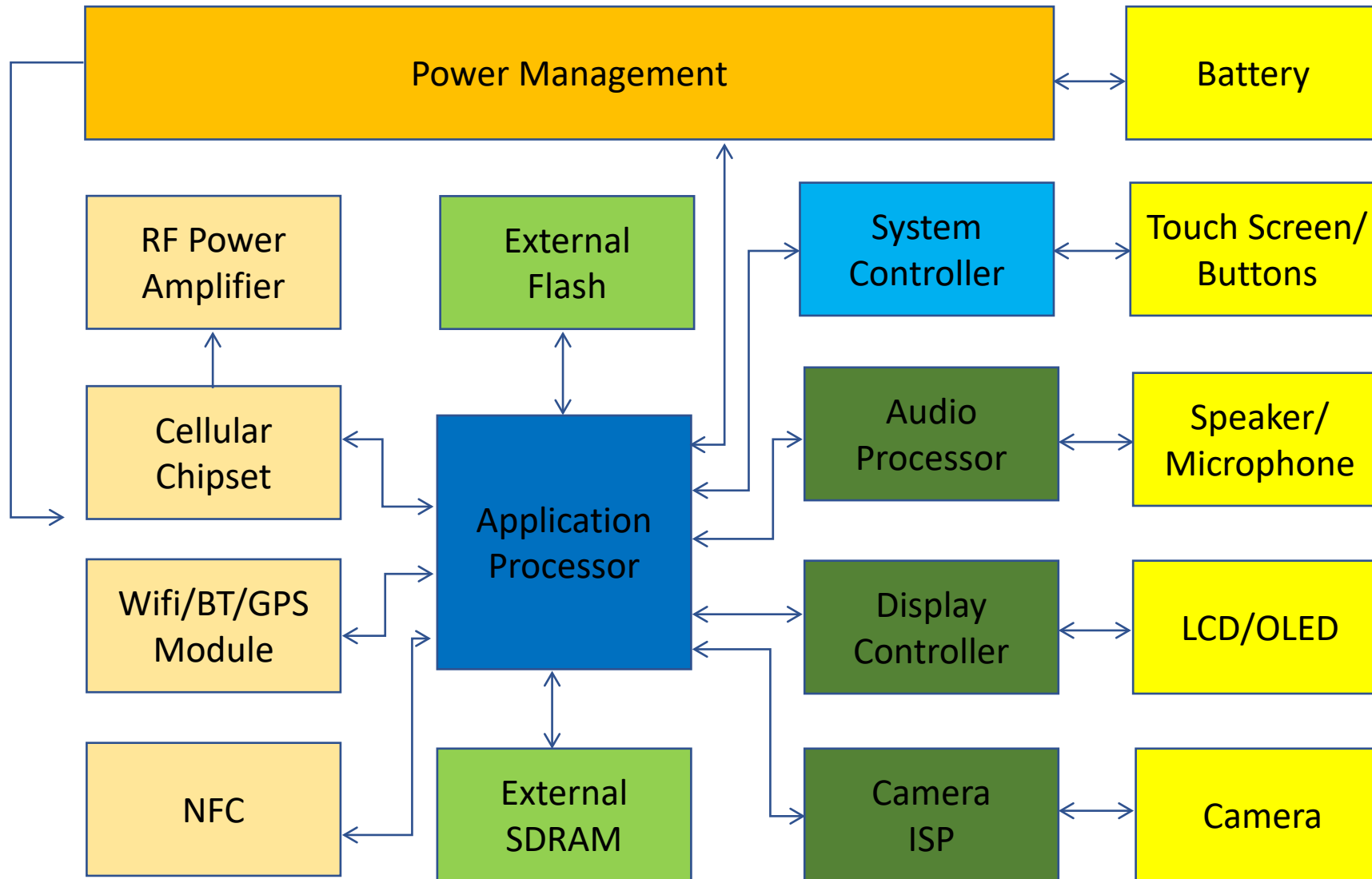


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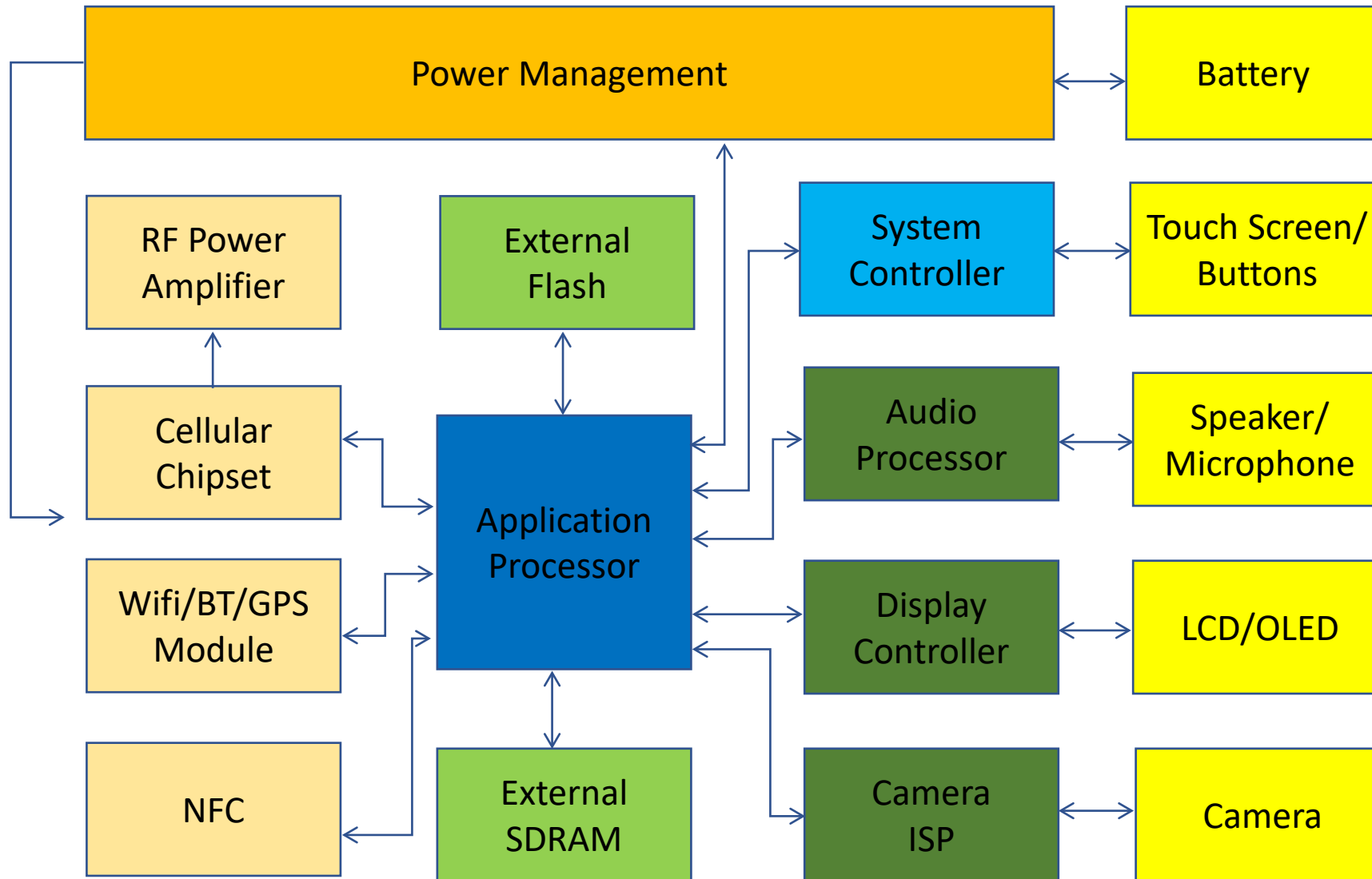
Computer System Block Diagram



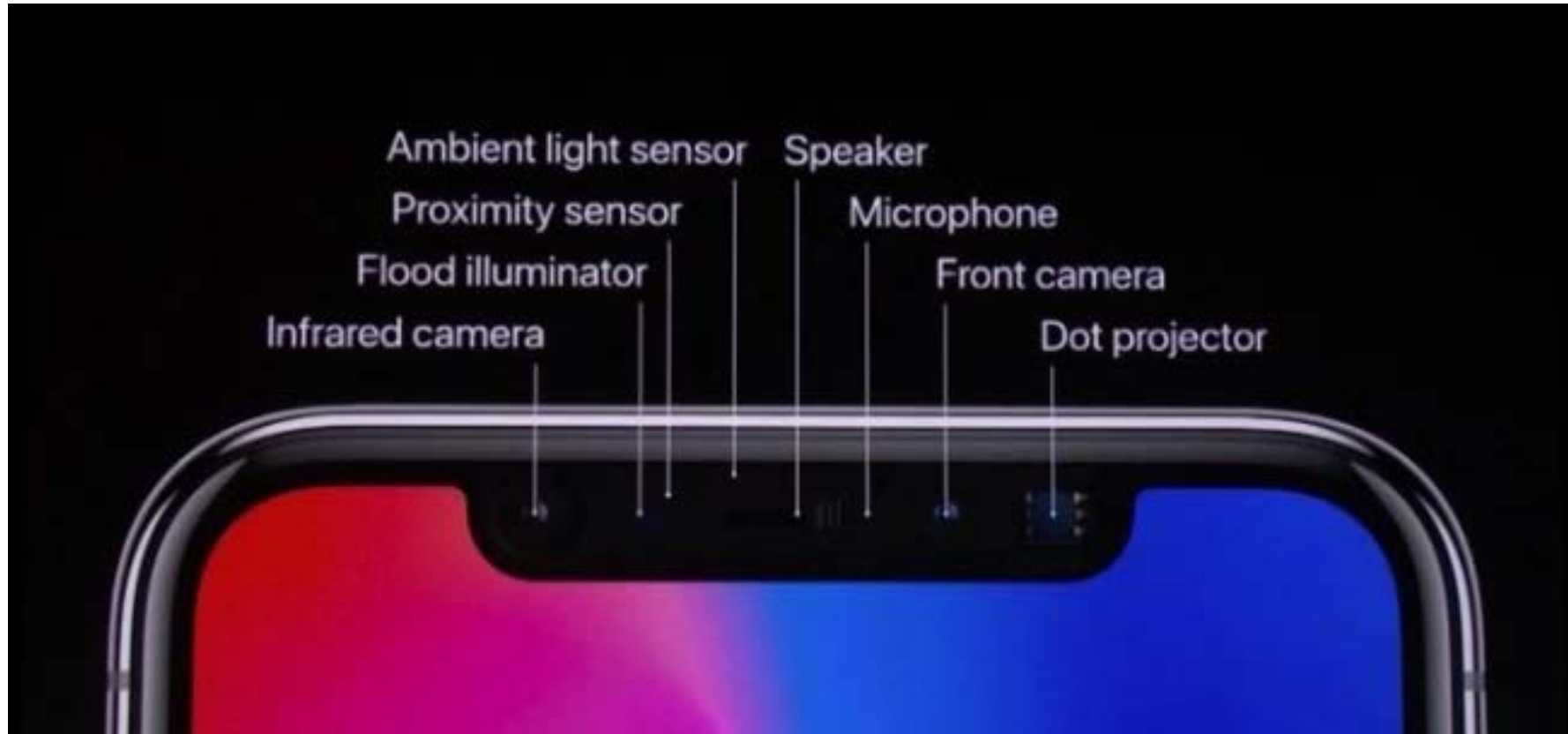
Example: Smart Phone



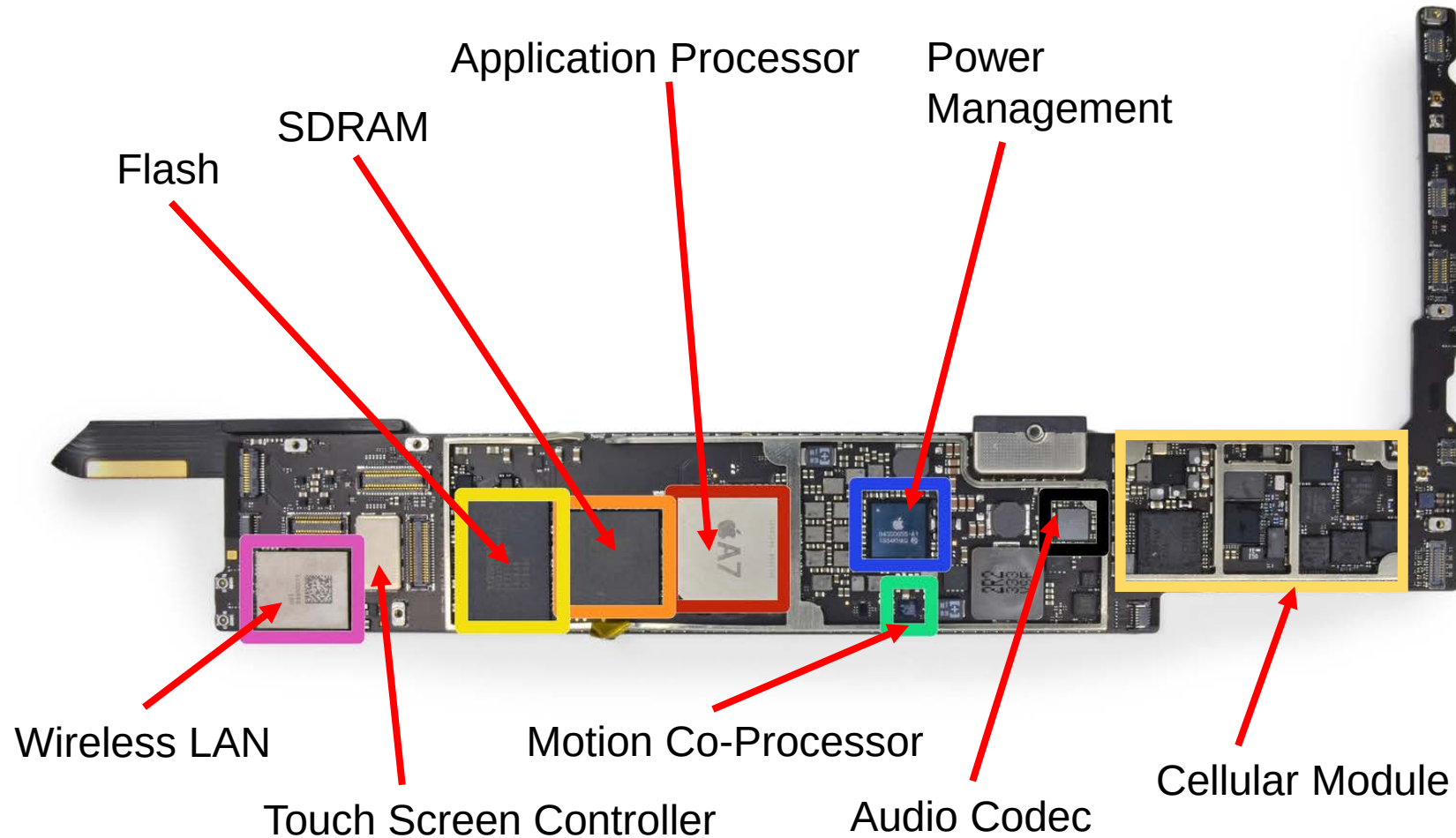
Example: Smart Phone



iPhone front facing peripherals



iPad Air Main PCB



Sub-Systems in a Computer System & Presentation Schedule in Part 2

- Signal Chain Sub-System
 - ADC-DAC
 - Parallel and Serial (UART/SPI) Digital Interfaces
 - Direct Memory Access Controller
 - Interrupt Controller
- Memory Sub-System
 - Semiconductor Memories (SRAM, DRAM, FLASH)
 - Flash memory based Solid State Drives
 - Magnetic Hard Disk Drives
 - Cache Memory Management
 - Virtual Memory Management
- Communication and User Interface Sub-System (brief intro only)
 - Wired (USB, SATA, HDMI)
 - Wireless (Wifi, Bluetooth)
 - Input Devices (KBM, Capacitive Touch, Camera, Microphone)
 - Output Devices (Display, Speakers)
- General data representation in a computer system
- More for Processor Core
 - Instruction Pipeline